

Legal AI

Legal AI answers questions about legal documents, such as statutes and contracts. It checks every sentence of its answer against the passage it cites before showing it. If the check fails, the system says it cannot answer instead of guessing.

This guide covers what makes the system different, what each page of the web app does, the step-by-step workflows behind them, and every algorithm used, with the file that implements it.

1. What makes it different

Most legal chatbots, including ones built on retrieval-augmented generation (RAG), fetch some passages and let the model write an answer. Nothing checks that answer before the user reads it. Legal AI adds that check and makes it visible.

	Typical RAG chatbot	Legal AI
Citations	Optional, often invented	Required on every claim; a citation to a passage that was not retrieved scores zero
Checking the answer	None	Four detectors check each claim against its source (V1–V4)
Deciding to answer	Always answers	Answers only if the Verification Confidence Score (VCS) is at least 0.60, otherwise abstains
Evidence for the user	A link at best	A proof for every claim: quoted passage, each check's verdict, the claim's score
General questions ("what is an NDA?")	Mixed in with sourced answers	Answered separately and labelled <i>general knowledge</i> , with no citations or score
Chunking	Fixed-size windows	Split by section and clause for statutes, by paragraph for anything else
Infrastructure	Vector database server	One embedded LanceDB folder on disk, with no server
Model lock-in	One provider	Claude, OpenAI, Gemini or Groq, chosen with one environment variable

What the measurements show (all logged under `/experiments`):

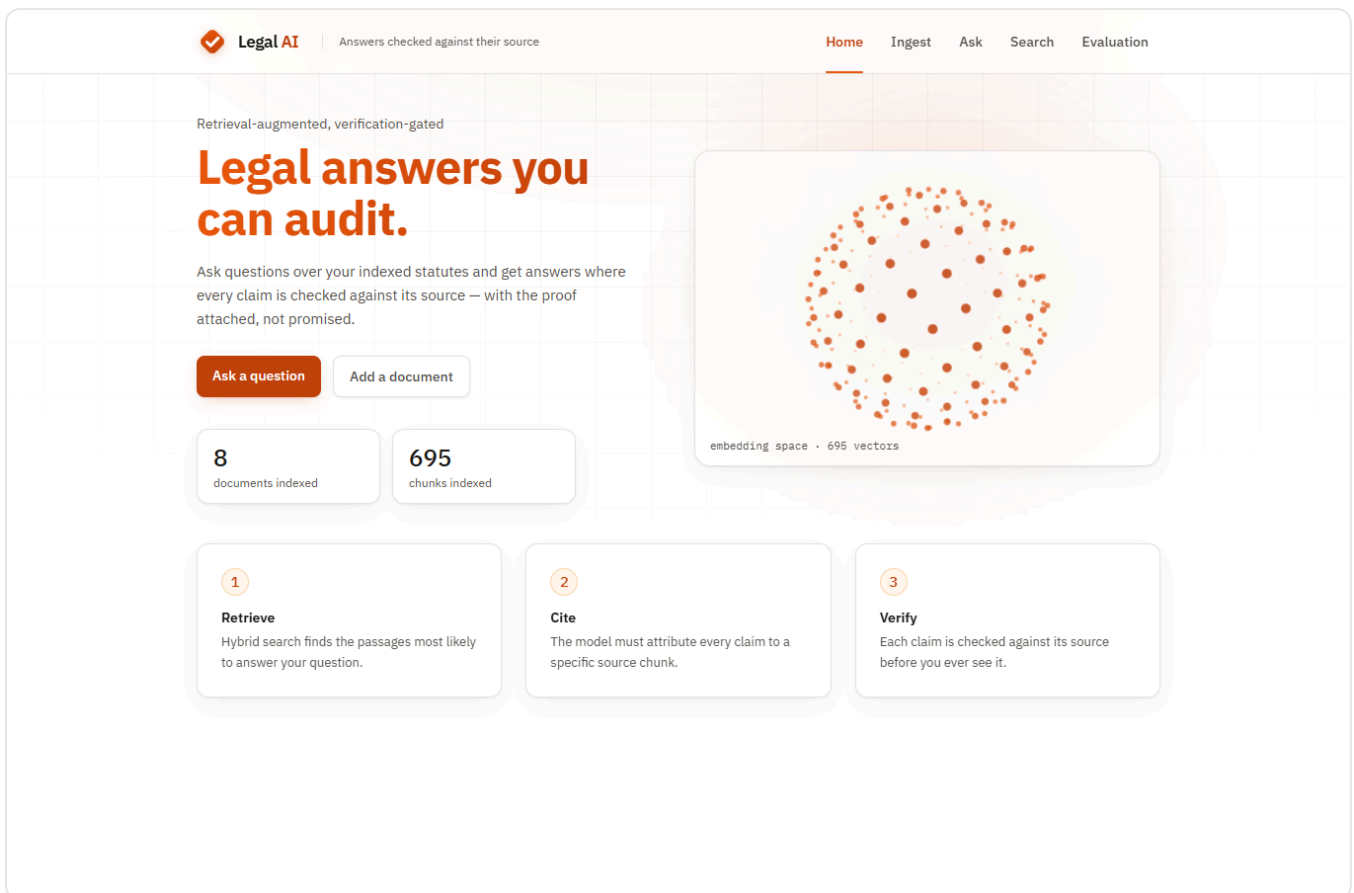
- We corrupted 28 claims on purpose, with fake citations and swapped claims. The full chain showed **none** of them to the user.
- Removing any single detector still showed none, because the detectors overlap. Removing the **score gate (V5)** showed **all 28**. The gate is what protects the user, not any one detector.

- Structure-aware chunking (SAC) raised retrieval F1 from **0.333 to 0.417** over paragraph chunking, and sent the model about a third of the text.

2. The pages

The web app (React + Vite + Tailwind) has five pages. Every page calls the FastAPI backend, and everything a page shows is also available as an API call.

2.1 Home



What you see:

- The headline and two buttons, *Ask a question* and *Add a document*.
- Two counters: documents indexed and chunks indexed.
- The three steps of the system: Retrieve, Cite, Verify.

Behind it:

- The counters come from `GET /stats`, which counts rows in the LanceDB table (`src/api/index_stats.py`).
- The rotating point cloud is decorative, drawn on a canvas with no 3D library. The vector count under it is the real chunk count.

2.2 Ingest

Ingest

Add documents to the index so they can be searched and cited.



Add a document to the index

Ingestion → summary-augmented chunking → embedding → LanceDB. Statutes chunk by section; other documents fall back to paragraph chunking.



Drop a statute here, or **browse**

.txt or .pdf — statutes chunk by section, anything else by paragraph

Ingest and index

What you see: a drop zone for a `.txt` or `.pdf` file and an *Ingest and index* button. After upload, the page shows:

- how many chunks were created;
- whether the file was split by legal structure (SAC) or fell back to paragraphs;
- the chunks themselves.

Behind it (`POST /ingest` , `src/api/routes/ingest.py`):

1. The file's bytes are hashed with SHA-256. If the same content or the same file name is already indexed, the server answers **409**. The page then offers **Replace existing document**, which calls `DELETE /documents/{source_id}` and uploads again.
2. The text is extracted (pypdf for PDF), cleaned and parsed for Act → Chapter → Section → Clause structure.
3. If there is structure, the document is split with summary-augmented chunking. Otherwise it is split by paragraph.
4. Each chunk is embedded and added to LanceDB, and the keyword index is rebuilt.

Each failure returns its own clear error:

Code	Cause
415	Unsupported file type
422	Empty file
503	Embedding model unavailable
500	Index write failed

2.3 Ask

The screenshot shows a user interface for asking questions. On the left, a sidebar titled 'Ask' contains a list of chat topics: '+ New chat', 'What penalty applies to a r...', 'why non disclosure agree...', and 'What happens if the Rece...'. The main area displays a chat conversation. The user's question is 'What happens if the Receiving Party breaches this agreement'. The system's response is a verified claim with a green badge: 'Verified VCS 0.62 · 2 claims checked'. The text of the claim is: 'If the Receiving Party breaches or threatens to breach the agreement, the Disclosing Party may obtain an injunction (final or interlocutory) to stop the breach, and it does not need to show any actual damage to do so.' This is followed by a citation chip: '09-24-2019-04-25-05-3914910473::p25' which 'contributes 1.00'. Below the citation is a quoted source passage: 'Breach In the event of a breach or threatened breach of the terms of this Agreement by the Receiving Party, the Disclosing Party will be entitled to seek an injunction (final or interlocutory) restraining the Receiving Party from committing a breach , or further breach, of this Agreement without the requirement to demonstrate any actual damage sustained by the Disclosing Party. 9. Integration This Agreement constitutes the entire understanding between the Parties concerning the treatment of information to which this Agreement relates. No modification of this Agreement shall be binding on either Party unless agreed in writing and signed by all the Parties. 10.' Below the quote are four verification checks: 'V1 citation exists', 'V2 support entails 0.88', 'V3 fidelity 100%', and 'V4 consistency skipped'. The text continues: 'The parties also agree that monetary damages alone are not an adequate remedy for a breach, and therefore each party is entitled to equitable relief such as injunctions, specific performance, or other equitable remedies for any threatened or actual breach.' This is followed by another citation chip: '5-NSK-Confidentiality-Agreement-for-Suppliers::p23'. At the bottom, there is a text input field 'Ask about a statute you've indexed...' and an 'Ask' button.

What you see:

- A chat on the right and saved conversations on the left.
- Each answer carries a badge:
 - **Verified:** green, with the VCS and the number of claims checked.
 - **Abstained:** amber.
 - **General knowledge:** violet.
- Every claim ends in a citation chip. Clicking the chip opens the **proof**:
 - the quoted source passage;
 - a chip for each check (V1 citation exists, V2 support, V3 fidelity, V4 consistency);
 - how much the claim contributes to the score.

Ask

Answers from your indexed documents, with every claim checked against its source.

+ New chat

What penalty applies to a ...

why non disclosure agree...

What happens if the Recei...

What penalty applies to a repeat unfair trade practice offence?



! Abstained VCS -

INSUFFICIENT_CONTEXT: The provided passages do not contain information about penalties for repeat unfair trade practice offences.



Abstained

The answer wasn't supported by the retrieved context, so nothing was asserted.

When the documents don't contain the answer, the model returns `INSUFFICIENT_CONTEXT` and the page states plainly that nothing was asserted.

Behind it:

- `POST /query` runs the whole pipeline: retrieval, generation, verification, and the general-knowledge fallback if needed (`src/api/routes/query.py`). The page sends `k = 12` , reranking on, and no self-consistency resamples (V4 shows as *skipped*, to save API calls).
- Conversations are saved as one JSON file each under `data/chats/` (`src/chat/store.py`) through `GET/POST/DELETE /chats` and `POST /chats/{id}/messages` .

2.4 Search

Search

Inspect what the retriever finds for a query, before any generation or verification.

Compare retrieval strategies
The same query run three ways — vector, keyword, and fused hybrid — so you can see where each finds the right passage.

☐ rerank

Dense - vector 5

Paragraph 22 0.748
09-24-2019-04-25-05-3914910473
Return of information and property The Receiving Party acknowledges and agrees that the Confidential Information is and remains the property of the Disclosing Party. The Receiving party must, at the end of this Agreement or within seven days following written instruction by the Disclosing Party to do so, return or destroy all Confidential Information in the Receiving Party's possession, custody or control.

Paragraph 23 0.755
09-24-2019-04-25-05-3914910473
The Receiving Party may retain one (1) copy of any such Confidential Information to be stored securely with its legal representative only if required for regulatory or insurance purposes, but in this event the confidentiality and non -use provisions in t his Agreement remain in place with respect to such retained Confidential Information. The return or destruction of any Confidential Information will not diminish or otherwise affect any other obligations under this Agreement. 6. No implied rights This Agreement shall not grant, nor be construed to grant any rights whatsoever beyond the scope of the Permitted Purpose of the Confidential Information to the Receiving Party .

Lexical - keyword 5

Paragraph 32 9.539
064-19 Non Disclosure Agreement 2019
c. was received by the Recipient without breach of this Agreement from a third party without restriction as to the use and disclosure of the DIVclRveU#IV Confidential Information; or,

Paragraph 25 9.378
09-24-2019-04-25-05-3914910473
Breach In the event of a breach or threatened breach of the terms of this Agreement by the Receiving Party, the Disclosing Party will be entitled to seek an injunction (final or interlocutory) restraining the Receiving Party from committing a breach , or further breach, of this Agreement without the requirement to demonstrate any actual damage sustained by the Disclosing Party. 9. Integration This Agreement constitutes the entire understanding between the Parties concerning the treatment of information to which this Agreement relates. No modification of this Agreement shall be binding on either Party unless agreed in writing and signed by all the Parties. 10.

Paragraph 18 9.881
354452_0000950123-99-011178_document_...
(a) is or subsequently becomes publicly available without the Receiving Party's breach of any obligation owed to the

Hybrid - fused 5

Paragraph 25 0.032
09-24-2019-04-25-05-3914910473
Breach In the event of a breach or threatened breach of the terms of this Agreement by the Receiving Party, the Disclosing Party will be entitled to seek an injunction (final or interlocutory) restraining the Receiving Party from committing a breach , or further breach, of this Agreement without the requirement to demonstrate any actual damage sustained by the Disclosing Party. 9. Integration This Agreement constitutes the entire understanding between the Parties concerning the treatment of information to which this Agreement relates. No modification of this Agreement shall be binding on either Party unless agreed in writing and signed by all the Parties. 10.

Paragraph 22 0.016
09-24-2019-04-25-05-3914910473
Return of information and property The Receiving Party acknowledges and agrees that the Confidential Information is and remains the property of the Disclosing Party. The Receiving party must, at the end of this Agreement or within seven days following written instruction by the Disclosing Party to do so, return or destroy all Confidential Information in the Receiving Party's possession, custody or control.

What you see: one query run three ways, side by side:

- **Dense** (vector);
- **Lexical** (keyword, BM25);
- **Hybrid** (the two fused).

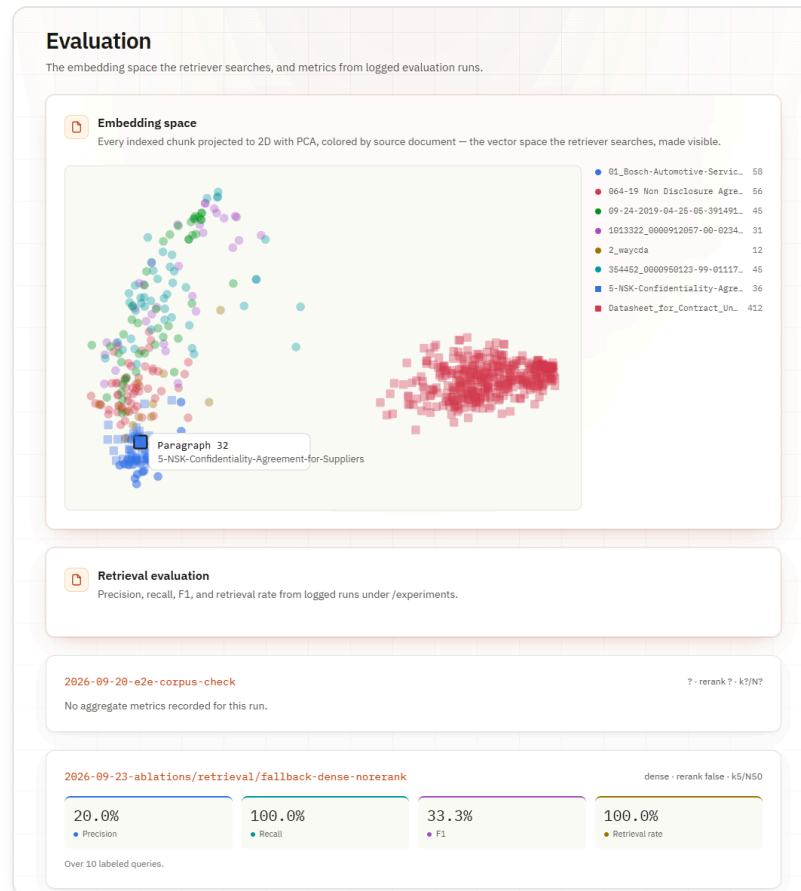
Each result shows its chunk, source document and score. A *rerank* checkbox adds a fourth column reranked by the cross-encoder.

How to read the scores:

- **Dense** shows vector distance. Lower means closer.
- **Lexical** shows a BM25 score. Higher is better.
- **Hybrid** shows the reciprocal rank fusion (RRF) score, which is higher when a chunk ranks well in *both* lists. In the screenshot, *Paragraph 25* appears in both the dense and keyword lists and scores 0.032; chunks found by only one list score about 0.016.

Behind it: `GET /retrieve?q=...&k=...&rerank=...` (`src/api/routes/retrieve.py`). No model generates anything here. The page exists to inspect retrieval on its own.

2.5 Evaluation



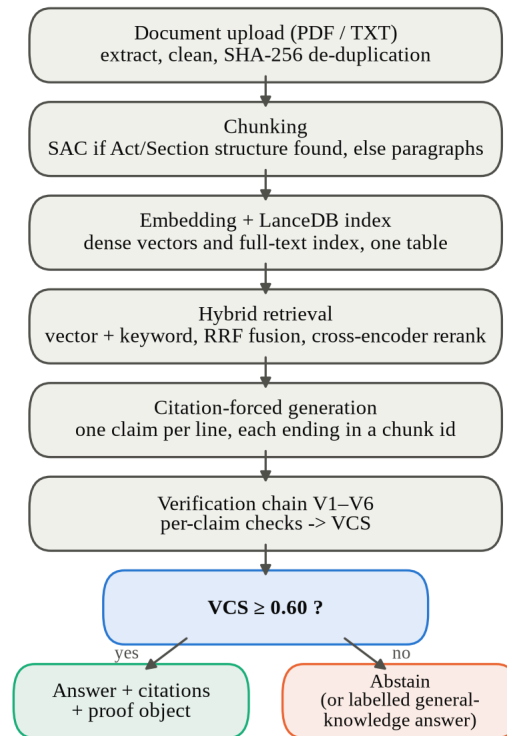
What you see:

1. **Embedding space.** Every indexed chunk is projected to 2D, with one colour and shape per document. Hovering a point shows which chunk it is. In the screenshot, the 412 chunks of the one non-agreement document form their own cluster, while the seven agreements overlap because they share vocabulary.
2. **Retrieval evaluation.** There is one card per logged experiment, with precision, recall, F1 and retrieval rate.

Behind it:

- `GET /embedding-map` loads every vector and projects it with PCA (`src/api/projection.py`).
- `GET /evaluation` walks `experiments/` and reads each run's `config.json` and `results.json` (`src/evaluation/results_store.py`).
- Verification-ablation runs store their results in a different format, so their cards currently say *No aggregate metrics*. Their numbers are in the paper and in `experiments/2026-09-23-ablations/verification/results.json` .

3. Workflows



3.1 Adding a document

Step	What happens	Code
1	Read the upload and hash it with SHA-256; refuse duplicates (409)	<code>src/indexing/dedup.py</code>
2	Extract text: plain read for <code>.txt</code> , pypdf for <code>.pdf</code>	<code>src/ingestion/loaders.py</code>
3	Clean whitespace without touching headings	<code>src/ingestion/cleaning.py</code>
4	Parse Act / Chapter / Section / Clause	<code>src/ingestion/structure.py</code>
5	Chunk: SAC if structure was found, else paragraph fallback	<code>src/chunking/sac.py</code> , <code>src/chunking/fallback.py</code>
6	Embed "summary + chunk text" with all-MiniLM-L6-v2	<code>src/indexing/build.py</code> , <code>src/embedding/</code>
7	Append rows to LanceDB and rebuild the keyword (FTS) index	<code>src/indexing/build.py</code>

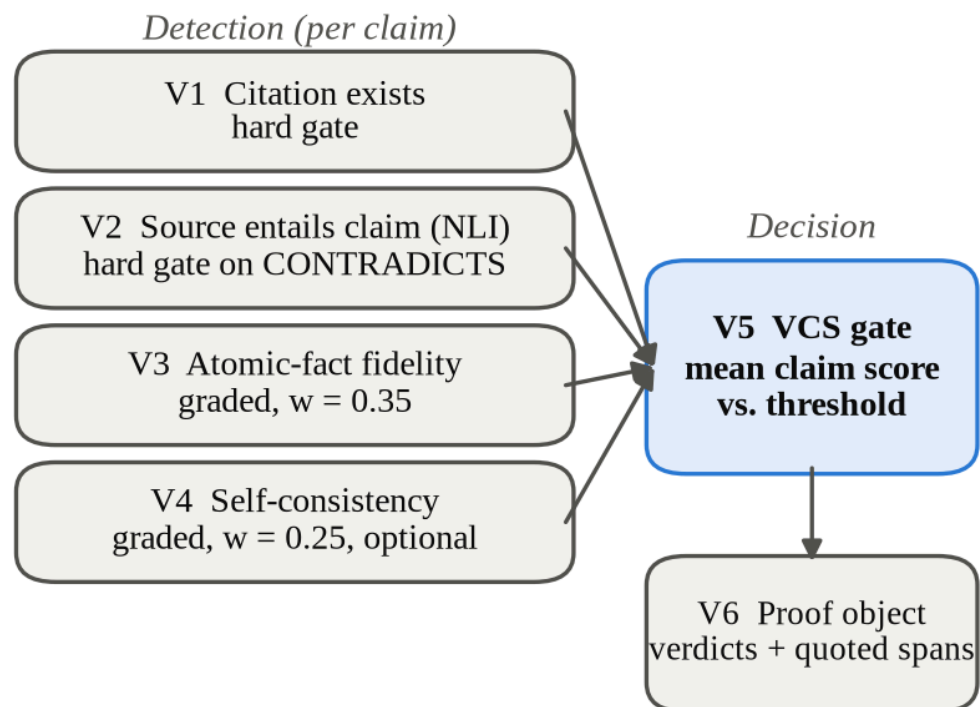
The file name (without extension, trimmed) becomes the `source_id` , and every chunk id starts with it:

- SAC ids look like `tenancy_act::s4:b` (Section 4, Clause b).
- Fallback ids look like `contract::p25` (Paragraph 25).

3.2 Asking a question

Step	What happens	Code
1	Fetch a deep candidate pool (at least 100) with hybrid search	<code>src/retrieval/retriever.py</code>
2	Rerank the pool with the cross-encoder and keep the top k (12 in the UI)	<code>src/retrieval/rerank.py</code>
3	Build the prompt: only these passages; one claim per line; every line ends in <code>[chunk_id]</code> ; or reply <code>INSUFFICIENT_CONTEXT</code>	<code>src/generation/prompt.py</code>
4	Call the chosen LLM through its adapter	<code>src/generation/adapters/</code> , <code>factory.py</code>
5	Parse the reply into claims and their cited ids	<code>src/generation/parser.py</code>
6	Run the verification chain V1–V6	<code>src/verification/chain.py</code>
7	If $VCS \geq 0.60$, the answer is Verified ; otherwise Abstained	<code>src/verification/v5_vcs.py</code>
8	If declined <i>and</i> the question is a general concept question, answer from model knowledge, labelled as unverified	<code>src/generation/general_knowledge.py</code>

3.3 Verifying an answer (V1–V6)



Layer	Question it answers	How	Effect
V1 Citation existence	Was the cited chunk actually retrieved?	Set membership against the retrieved ids	Fail → claim score 0 (hard gate)

Layer	Question it answers	How	Effect
V2 Entailment	Does the cited passage support the claim?	NLI with roberta-large-mnli: passage = premise, claim = hypothesis	ENTAILS → 1, NEUTRAL → 0, CONTRADICTS → claim score 0 (hard gate)
V3 Atomic fidelity	Is every part of the claim supported?	Split the claim into small facts; NLI on each; fraction entailed	0 to 1, weight 0.35
V4 Self-consistency	Does the claim come back if we ask again?	Regenerate N times; match claims by word overlap	0 to 1, weight 0.25 (optional)
V5 VCS gate	Is the whole answer trustworthy enough?	Weighted score per claim, averaged, compared to $\tau = 0.60$	ANSWER or ABSTAIN
V6 Proof object	What is the evidence?	Records each claim, cited ids, quoted passage, verdicts, score	Shown when a citation is clicked

Claim score:

```
if V1 failed or V2 = CONTRADICTS:    s = 0
else:  s = (0.40·e + 0.35·f + 0.25·c) / (sum of the weights of the layers that ran)
```

e = 1 if V2 says ENTAILS, else 0; **f** = V3 fidelity; **c** = V4 consistency. A layer that did not run is removed from both the top and the bottom of the fraction, so skipping V4 never lowers a score.

Answer score: VCS = the average of the claim scores. The answer is shown if VCS ≥ 0.60 .

Worked example (the Ask screenshot):

- The first claim has V1 exists, V2 entails, V3 fidelity 100% and V4 skipped. Its score is $(0.40 \cdot 1 + 0.35 \cdot 1) / (0.40 + 0.35) = \mathbf{1.00}$.
- The answer's VCS is 0.62 across two claims, so the second claim scored below 0.25. It was shown because it was averaged with a strong claim. This is a known weakness of mean aggregation (see §7).

3.4 Searching

`GET /retrieve` runs dense, keyword and hybrid search for the same query and returns all of them; with `rerank=true` it adds the reranked hybrid list. It never calls the LLM.

3.5 Evaluating and ablations

Script	What it does
<code>python -m src.evaluation.run_retrieval_eval</code>	Runs labelled questions and scores precision, recall, F1 and retrieval rate. Writes <code>experiments/<run>/config.json</code> and <code>results.json</code>
<code>python -m src.evaluation.run_ablation_sweep</code>	Every combination of chunking (SAC / fallback), retrieval (dense / hybrid) and reranking (on / off)
<code>python -m src.evaluation.run_verification_ablation</code>	Generates each answer once, then replays it with one layer switched off at a time. Also runs two probes: fake citations and swapped claims

Script	What it does
<code>python docs/paper/make_figures.py</code>	Redraws every paper figure from the logged files

3.6 Managing documents and chats

- **Duplicate check:** same bytes (SHA-256) or same name → 409, and nothing is added.
- **Replace:** delete the old document's chunks, then upload again.
- **Delete:** `DELETE /documents/{source_id}` removes every chunk of that document and rebuilds the keyword index, so deleted text can't still turn up in keyword search.
- **Chats:** stored as one JSON file per conversation. Conversation ids are checked, so a crafted id can't point outside the chats folder.

4. Algorithms

#	Algorithm	Where	How it works
1	Text cleaning	<code>ingestion/cleaning.py</code>	Normalise line endings, strip trailing spaces, collapse 3+ blank lines. Headings are left intact
2	Structure parsing	<code>ingestion/structure.py</code>	A line-by-line regex parser recognising <code>CHAPTER IV</code> , <code>Section 5. ...</code> , <code>(1) ...</code> and <code>Clause (a):</code>
3	Summary-Augmented Chunking (SAC)	<code>chunking/sac.py</code>	One chunk per section body and one per clause. Each carries a short summary of its section (heading + first sentence), made without an LLM. The summary is embedded with the chunk, so a clause is found together with its section's context
4	Paragraph fallback chunking	<code>chunking/fallback.py</code>	Split on blank lines. A paragraph over 700 characters is split into sentences, and the sentences are packed greedily into windows of at most 700 characters
5	SHA-256 fingerprinting	<code>indexing/dedup.py</code>	A hash of the raw bytes is stored on every chunk; a matching hash or name means a duplicate
6	Sentence embeddings	<code>embedding/sentence_transformer.py</code>	all-MiniLM-L6-v2, 384 dimensions, normalised to unit length
7	BM25 keyword search	LanceDB full-text index	Scores chunks by term frequency, weighted by how rare each term is, with document-length normalisation
8	Reciprocal Rank Fusion (RRF)	<code>indexing/query.py</code> (LanceDB)	$\text{score}(d) = \sum \frac{1}{(60 + \text{rank of } d \text{ in each list})}$. Rewards chunks that rank well in both dense and keyword search
9	Cross-encoder reranking	<code>retrieval/rerank.py</code>	BAAI/bge-reranker-base reads query and passage <i>together</i> and gives a relevance score. Applied to the top 100 candidates, then cut to <i>k</i>

#	Algorithm	Where	How it works
10	Neighbour expansion	<code>retrieval/neighbors.py</code>	Adds the chunks just before and after each hit from the same document, for answers split across chunks. Off by default
11	Citation-forced prompting	<code>generation/prompt.py</code>	System rules: use only the context, one claim per line, cite every line, or reply <code>INSUFFICIENT_CONTEXT</code>
12	Claim parsing	<code>generation/parser.py</code>	Regex pulls <code>[id]</code> , <code>[id1][id2]</code> or <code>[id1, id2]</code> from each line; a line with no citation is flagged as malformed
13	Citation existence (V1)	<code>verification/v1_citation.py</code>	Every cited id must be in the set of retrieved ids
14	Natural language inference (V2)	<code>verification/nli.py</code> , <code>v2_entailment.py</code>	roberta-large-mnli labels a (passage, claim) pair ENTAILS / NEUTRAL / CONTRADICTS. With several citations, the best outcome is kept (ENTAILS beats CONTRADICTS beats NEUTRAL)
15	Atomic decomposition (V3)	<code>verification/v3_atomic.py</code>	Split the claim on <code>;</code> , <code>"</code> and <code>,</code> and sentence ends; check each part with NLI; fidelity = parts entailed ÷ parts
16	Self-consistency (V4)	<code>verification/v4_consistency.py</code>	Regenerate N answers. A claim "recurs" if a resampled claim shares ≥ 50% of its words (token Jaccard). Consistency = recurrences ÷ N
17	Verification Confidence Score (V5)	<code>verification/v5_vcs.py</code>	Hard gates, then the weighted mean per claim, the average over claims, and the threshold 0.60 (see §3.3)
18	Proof object (V6)	<code>verification/v6_proof.py</code>	For each claim, picks the cited chunk with the most word overlap as the quoted passage and stores every verdict
19	General-concept detection	<code>generation/general_knowledge.py</code>	Regex. The question must start like a definition ("what is", "define", "explain") <i>and</i> have no reference to the documents ("this agreement", "section 4", "what happens if"). Unclear cases stay on the verified path
20	PCA projection	<code>api/projection.py</code>	Centre the vectors, take the top 2 singular vectors (NumPy SVD), project onto them
21	Retrieval metrics	<code>evaluation/retrieval_eval.py</code>	$P = \text{relevant retrieved} \div \text{retrieved}$; $R = \text{relevant retrieved} \div \text{relevant}$; $F1 = 2PR / (P + R)$; retrieval rate = share of questions with at least one gold chunk in the top k

5. API

Method	Path	Purpose
POST	/ingest	Upload a <code>.txt</code> / <code>.pdf</code> , chunk, embed, index
DELETE	/documents/{source_id}	Remove a document's chunks
GET	/retrieve	Dense / keyword / hybrid (/ reranked) results for a query
POST	/query	Full pipeline: answer + VCS + proof
GET	/stats	Documents and chunks indexed
GET	/embedding-map	2D PCA coordinates of every chunk
GET	/evaluation	Logged experiment runs
GET POST DELETE	/chats, /chats/{id}, /chats/{id}/messages	Conversation history
GET	/health	Liveness check

6. Technology

Part	Choice
Backend	Python, FastAPI
Index	LanceDB (embedded; vectors + BM25 in one on-disk table)
Embeddings	sentence-transformers/all-MiniLM-L6-v2
Reranker	BAAI/bge-reranker-base (cross-encoder)
Entailment	roberta-large-mnli (Hugging Face Transformers)
Generation	Claude, OpenAI, Gemini or Groq via adapters (<code>LLM_PROVIDER</code>)
PDF	pypdf
Frontend	React, Vite, Tailwind CSS, Framer Motion
Tests	pytest (unit tests use stub models, so no downloads or API keys are needed)

7. Limitations

- **Small test set:** ten labelled questions over five synthetic statutes. Every retrieval setting found a gold clause for every question, so the set can't rank retrieval methods.

- **Threshold not tuned:** $\tau = 0.60$ is a default. Three of ten answerable questions were refused because the NLI model said NEUTRAL on correct citations.
- **Mean aggregation:** a weak claim can pass next to a strong one (the §3.3 example). Gating each claim separately would fix this.
- **General-domain NLI:** roberta-large-mnli was not trained on legal text.
- **Missing pieces:**
 - HTML files can't be ingested yet.
 - The Evaluation page doesn't show verification-ablation runs.
 - Reranking adds several seconds per question.